SIMATS ENGINEERING

ASSESSMENT TOOL 3

Course: Internet Programming (CSA43) | CO-1 | Total Marks: 10

Field	Details
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Course Name / Code	Internet Programming / CSA43
Course Outcome	CO-1: Develop standards-compliant web pages using HTML/XHTML and XML; apply Internet protocols and HTTP (L2)
Scenario	Web Development Standards Implementation

Understanding of the Case

A start-up is building a modern web application and has run into five problems. Each one comes from not following a web standard. The table below states each issue, the standard that applies, and the fix chosen in this report.

Observation	Underlying issue	Solution chosen
Promotional video on the homepage	Using the right HTML5 media element	<video> with controls, <source> and a fallback
Registration form must send data securely	Choosing the right HTTP method and transport	POST over HTTPS, with server-side validation
Navigation menu	Semantic markup for accessibility and SEO	<nav> containing a list of links
XML configuration errors	Well-formedness rules	Fix: one root, closed tags, proper nesting, complete tags
Correct web protocol	Application-layer protocol for the web	HTTP, upgraded to HTTPS in production

All five solutions work together in the integrated example given after Answer 5.

Answer 1: Video Embedding

Issue

The homepage needs a promotional video. Older sites used plugins such as Flash or <embed>, which are now unsupported, insecure and unavailable on mobile browsers. HTML5 solved this with a native element.

Correct option: B. `<video src="video.mp4" controls></video>`

Justification

- **Standard element:** <video> was introduced in HTML5 and is supported by all modern browsers, so no plugin is needed.
- **User control:** the controls attribute shows the browser's play/pause, seek, volume and fullscreen buttons. Without it the video has no visible controls.

- **Invalid options:** <media>, <movie> and <vid> (A, C, D) are not defined in the HTML specification. A browser treats them as unknown inline elements and shows nothing, so the video never plays.

Useful attributes of <video>

Attribute	Purpose
`controls`	Shows the built-in playback controls
`poster`	Image shown before the video starts, which also helps the page look complete while loading
`preload`	none / metadata / auto: controls how much of the file is fetched in advance. metadata saves bandwidth
`autoplay + muted`	Browsers block autoplay with sound, so autoplay only works when the video is muted
`loop`	Restarts the video when it ends
`width / height`	Reserve space so the layout does not jump while loading

Solution design (improved code)

```
<video width="640" height="360" controls preload="metadata" poster="thumbnail.jpg">
  <source src="promo.mp4"  type="video/mp4">
  <source src="promo.webm" type="video/webm">
  <track kind="captions" src="captions_en.vtt" srclang="en" label="English">
  Your browser does not support the video tag.
</video>
```

- **Multiple <source> tags:** the browser plays the first format it supports (MP4/H.264 is widely supported, WebM is a good open alternative).
- **<track>:** adds captions, which helps deaf users and viewers who watch without sound, and meets accessibility guidelines.
- **Fallback text:** shown only by very old browsers that do not understand <video>.
- **Responsive tip:** add CSS video { max-width: 100%; height: auto; } so the video fits phone screens.

Answer 2: HTTP Method for Form Submission

Issue

A registration form sends a name, e-mail and password to the server. This data is private and the request creates a new user record, so the method must keep data out of the URL and must suit data submission.

Correct option: B. POST

Justification

- **Security:** POST sends data in the request body. It does not appear in the address bar, browser history, bookmarks, or the Referer header. With GET, a password would appear in the URL (for example /register?password=1234) and be stored in server logs.
- **Size:** URLs are limited (browsers and servers commonly cap them at a few thousand characters). The POST body has no such practical limit, so large forms and file uploads work.
- **Meaning of the method:** GET is a safe, repeatable read. POST is meant for actions that change server state, such as creating an account. Browsers warn before re-sending a POST, which prevents accidental duplicate registrations.

Comparison of all four options

Method	Purpose	Data location	Suitable for registration?
GET	Retrieve a resource	URL query string	No: data is exposed in the URL
POST	Submit or create data	Request body	Yes
PUT	Replace or update a resource	Request body	No: HTML forms cannot send PUT; used in APIs
DELETE	Remove a resource	Usually none	No: it deletes data

Note: HTML <form> supports only GET and POST. PUT and DELETE are used through JavaScript (fetch/AJAX) or REST APIs.

Solution design

```
<form action="/register" method="POST">
  <label for="user">Username</label>
  <input type="text" id="user" name="username" required>

  <label for="email">Email</label>
  <input type="email" id="email" name="email" required>

  <label for="pwd">Password</label>
  <input type="password" id="pwd" name="password" minlength="8" required>

  <button type="submit">Register</button>
</form>
```

POST alone does not encrypt anything. To send user data securely the application should also:

- Serve the page and the form action over HTTPS so the body is encrypted in transit.
- Validate all input on the server, because client-side checks (required, minlength) can be bypassed.
- Store passwords as salted hashes (for example bcrypt), never as plain text.
- Add a CSRF token to the form to stop forged cross-site submissions.
- Use enctype="multipart/form-data" only when the form uploads files.

Answer 3: Semantic Element for Navigation

Issue

Building the menu from tags works visually but gives the browser, screen readers and search engines no information about its purpose. Semantic HTML5 elements describe the role of each part of the page.

Correct option: B. <nav>

Justification

- **Meaning:** <nav> marks a section of major navigation links, such as the main menu or a table of contents.
- **Accessibility:** screen readers announce it as a navigation landmark and let users jump straight to it or skip it.
- **SEO:** search engines can identify the site structure and the important internal links.

- **Maintainability:** <nav> is easier to read and style than a series of nested tags.

Why the other options are not correct

Element	Real purpose
`<div>`	Generic container with no meaning. It is a fallback when no semantic element fits.
`<section>`	Thematic grouping of content, usually with a heading (for example 'Features'). It does not mean navigation.
`<header>`	Introductory content such as the logo, title and often the menu. It can contain a <nav>, but it is not the navigation element.

Solution design

```
<header>
  <h1>My Start-up</h1>
  <nav aria-label="Main navigation">
    <ul>
      <li><a href="index.html" aria-current="page">Home</a></li>
      <li><a href="about.html">About</a></li>
      <li><a href="register.html">Register</a></li>
      <li><a href="contact.html">Contact</a></li>
    </ul>
  </nav>
</header>
```

- A <ul> list inside <nav> tells assistive technology how many links there are.
- aria-label separates two navigation blocks on one page (for example main menu and footer menu).
- aria-current="page" marks the page the visitor is on.
- Use <nav> only for major navigation blocks, not for every group of links.

Other semantic elements to complete the page layout: <main>, <article>, <aside>, <footer>.

Simple CSS for a horizontal menu:

```
nav ul { list-style: none; display: flex; gap: 1rem; padding: 0; }
nav a  { text-decoration: none; color: #1F3864; }
nav a:hover, nav a[aria-current="page"] { text-decoration: underline; }
```

Answer 4: Correct XML Syntax

Issue

The configuration files contain syntax errors, so an XML parser will reject them. XML is strict: a document is usable only if it is well-formed. Unlike HTML, browsers do not correct XML mistakes.

Correct option: B. `<note><to>User</to><from>Admin</from></note>`

Justification

Option B is well-formed: the single root <note> contains <to> and <from>, each properly opened and closed, and everything nests correctly.

Errors in the other options and their corrections

Option	Code	Error	Corrected version

A	`<note><to>User</to><from>Admin</from>`	Root <note> is never closed (missing </note>)	`...<from>Admin</from></note>`
C	`<note><to>User</from></note>`	Mismatched tags: <to> is closed by </from>, and <from> was never opened	`<note><to>User</to></note>`
D	`<note to="User"></note`	Closing tag is incomplete: the > is missing	`<note to="User"></note>` or `<note to="User"/>`

Rules of well-formed XML

- Exactly one root element contains all others.
- Every start tag has a matching end tag (or is self-closing, like <item/>).
- Elements must nest correctly; they cannot overlap.
- Tags are case-sensitive: <Note> and <note> are different.
- Attribute values must be quoted, and an attribute name cannot repeat in one tag.
- Reserved characters must be escaped: < for <, & for &, > for >, " and ' inside attributes.
- Element names cannot start with a number or contain spaces.

Well-formed vs valid

A well-formed document follows the syntax rules above. A valid document is well-formed and also follows a DTD or XML Schema (XSD) that defines which elements are allowed. A configuration file should be both.

Solution design: a corrected configuration file

```
<?xml version="1.0" encoding="UTF-8"?>
<config>
  <server host="localhost" port="8080"/>
  <database>
    <name>appdb</name>
    <user>admin</user>
  </database>
  <!-- Use &amp; for the ampersand character -->
  <title>Sales &amp; Marketing</title>
</config>
```

Tip: opening the file in a browser, or running xmllint --noout config.xml, immediately reports the line number of any error.

Answer 5: Web Communication Protocol

Issue

The application needs a protocol that lets browsers request pages, media and data from a server and receive responses. The protocol must be the standard that every browser and web server understands.

Correct option: B. HTTP

Justification

- **Purpose:** HTTP (HyperText Transfer Protocol) is the application-layer protocol of the World Wide Web. It transfers HTML, CSS, JavaScript, images, video and API data.

- **Model:** client-server, request-response. The browser sends a request (method, URL, headers, optional body) and the server returns a response (status code, headers, body).
- **Stateless:** each request is independent. Applications keep a user logged in using cookies or tokens.
- **Transport:** runs over TCP (HTTP/1.1 and HTTP/2) or over QUIC/UDP (HTTP/3). Default port 80.
- **Secure version:** HTTPS is HTTP inside TLS encryption (port 443). It protects data from eavesdropping and tampering and confirms the server's identity. It is required for forms that carry passwords.

A request and response

```
GET /index.html HTTP/1.1
Host: www.example.com

HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Content-Length: 1256
```

Common status codes

Code	Meaning
`200 OK`	Request succeeded
`201 Created`	A new resource was created (typical after a successful registration)
`301 / 302`	Redirect to another URL
`400 / 404`	Bad request / page not found
`500`	Server error

Why the other protocols are not suitable

Protocol	Full form	Port	Use
FTP	File Transfer Protocol	21	Uploading and downloading files between hosts. Browsers do not use it to render pages.
SMTP	Simple Mail Transfer Protocol	25 / 587	Sending e-mail between mail servers
SNMP	Simple Network Management Protocol	161	Monitoring and managing network devices such as routers
HTTP	HyperText Transfer Protocol	80	Web pages and web APIs (correct choice)
HTTPS	HTTP over TLS	443	Secure version of HTTP, recommended for production

Integrated Solution

The following page combines all five solutions: semantic navigation, an embedded video, and a secure registration form sent with POST over HTTPS. The configuration the server reads is the well-formed XML shown in Answer 4.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>My Start-up</title>
</head>
<body>
  <header>
    <h1>My Start-up</h1>
    <nav aria-label="Main navigation">
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="#register">Register</a></li>
      </ul>
    </nav>
  </header>
  <main>
    <section>
      <h2>Watch our story</h2>
      <video width="640" height="360" controls poster="thumbnail.jpg">
        <source src="promo.mp4" type="video/mp4">
        Your browser does not support the video tag.
      </video>
    </section>
    <section id="register">
      <h2>Register</h2>
      <form action="https://www.example.com/register" method="POST">
        <input type="text" name="username" placeholder="Username" required>
        <input type="password" name="password" minlength="8" required>
        <button type="submit">Register</button>
      </form>
    </section>
  </main>
  <footer><p>&copy; 2026 My Start-up</p></footer>
</body>
</html>
```

Summary

Q	Answer	Key reason
1	`<video src="video.mp4" controls>`	Valid HTML5 native video element; others are not real elements
2	`POST`	Data goes in the request body, no size limit, used to submit data; use with HTTPS
3	`<nav>`	Semantic landmark: accessible, SEO-friendly, easy to maintain
4	`Option B`	Well-formed: single root, closed tags, correct nesting. A, C, D break these rules
5	`HTTP`	Standard web client-server protocol; use HTTPS in production

Conclusion: following these standards makes the application work in every browser, keeps user data safer, improves accessibility and search ranking, and prevents configuration failures.

Code of Conduct

I **T. LAKSHMI PRASAD NAIDU (Reg. No: 192324187)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
Total	100				